

Cryptology Assignment-4

Decrypt the following ciphertext using HILL cipher.

1. Cipher Text: HLDTOIHJDXWQCMAG

$$\text{Key} = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$$

Use (A=0, B=1, ..., Z=25)

Find the Plain Text.

Ans: $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \Rightarrow 21 - 10 = 11$

$\Rightarrow 11$ is a coprime with 26.

$$\Rightarrow 11^{-1} = 19 \pmod{26}$$

$$K^{-1} = 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \pmod{26} \Rightarrow \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

$$HL = [7, 11] = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 11 \end{bmatrix} \pmod{26} = \begin{bmatrix} 21 + 154 \\ 63 + 55 \end{bmatrix} \pmod{26} = \begin{bmatrix} 175 \\ 118 \end{bmatrix} \pmod{26} = \begin{bmatrix} 19 \\ 4 \end{bmatrix} \Rightarrow \begin{matrix} T \\ O \end{matrix}$$

$$DT = [3, 19] = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 19 \end{bmatrix} \pmod{26} = \begin{bmatrix} 15 \\ 18 \end{bmatrix} \Rightarrow \begin{matrix} P \\ S \end{matrix}$$

$$QI = [16, 8] = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 16 \\ 8 \end{bmatrix} \pmod{26} = \begin{bmatrix} 4 \\ 2 \end{bmatrix} \Rightarrow \begin{matrix} E \\ C \end{matrix}$$

$$HJ = [7, 9] = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 9 \end{bmatrix} \pmod{26} = \begin{bmatrix} 17 \\ 4 \end{bmatrix} \Rightarrow \begin{matrix} R \\ E \end{matrix}$$

$$DX = [3, 23] = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 23 \end{bmatrix} \pmod{26} = \begin{bmatrix} 19 \\ 12 \end{bmatrix} \Rightarrow \begin{matrix} T \\ M \end{matrix}$$

$$WQ = [22, 16] = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 22 \\ 16 \end{bmatrix} \pmod{26} = \begin{bmatrix} 4 \\ 18 \end{bmatrix} \Rightarrow \begin{matrix} E \\ S \end{matrix}$$

$$CM = [2, 12] = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 2 \\ 12 \end{bmatrix} \pmod{26} = \begin{bmatrix} 18 \\ 0 \end{bmatrix} \Rightarrow \begin{matrix} S \\ A \end{matrix}$$

$$AG = [0, 6] = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 0 \\ 6 \end{bmatrix} \pmod{26} = \begin{bmatrix} 8 \\ 4 \end{bmatrix} \Rightarrow \begin{matrix} G \\ E \end{matrix}$$

A 0	Q 16
B 1	R 17
C 2	S 18
D 3	T 19
E 4	U 20
F 5	V 21
G 6	W 22
H 7	X 23
I 8	Y 24
J 9	Z 25
K 10	
L 11	
M 12	
N 13	
O 14	
P 15	

Plain Text: TOP SECRET MESSAGE.

2. Consider the HILL Cipher Key Matrix

$$K = \begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix}$$

Cipher Text: RXAODY

Find the plain text for the above cipher text.

Ans: Decryption,

$$K^{-1} = C \cdot \text{mod } 26$$

$$\begin{aligned} \text{Det}(K) &= (24 \times 14) - (19 \times 5) \\ &= 336 - 95 = 241 \equiv 7 \end{aligned}$$

$\Rightarrow 7$ is a coprime with 26.

$$\Rightarrow 7^{-1} = 15$$

$$\Rightarrow K^{-1} = 15 \begin{bmatrix} 14 & -19 \\ -5 & 24 \end{bmatrix} = \begin{bmatrix} 210 & 105 \\ 315 & 360 \end{bmatrix} \text{ mod } 26$$

$$K^{-1} = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix}$$

$$RX = \begin{bmatrix} 17 \\ 23 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 17 \\ 23 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 57 \\ 557 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix} \Rightarrow \begin{matrix} F \\ L \end{matrix}$$

$$AO = \begin{bmatrix} 0 \\ 14 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 0 \\ 14 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 14 \\ 22 \end{bmatrix} \Rightarrow \begin{matrix} O \\ W \end{matrix}$$

$$DY = \begin{bmatrix} 3 \\ 24 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 3 \\ 24 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 84 \\ 17 \end{bmatrix} \Rightarrow \begin{matrix} E \\ R \end{matrix}$$

Plain Text: FLOWER FLOWER.

3. Encrypt "HI" using the Affine

3. Confusion v/s Diffusion

Analyze the following classical ciphers with respect to confusion and diffusion. Complete the table and justify your classification based on how each cipher transforms the plaintext.

Ans:

Cipher	Does it change the character/ symbol value?	Does it change the character position?	Confusion/ Diffusion	Justification
Shift Cipher	Yes	No	Confusion	Substituted via mathematical shift ($P+K$); character order is unaltered.
Multiplicative Cipher	Yes	No	Confusion	Substituted via modular multiplication ($P \times K$); character order is unaltered.
Transposition Cipher	No	Yes	Diffusion	Characters retain original value, but their positions are scrambled across the text.
Affine Cipher	Yes	No	Confusion	Combines Multiplicative & Shift structures ($(a \cdot P + b) \bmod 26$); position remains unchanged.

Relationship between Substitution & Confusion

In a substitution cipher, the position remains the same, but the actual character value changes.

$A \rightarrow D$

$\therefore$ The relationship between the plaintext characters and ciphertext character is hidden. Hence substitution mainly provides confusion.

Relationship between Transposition & Diffusion

In transposition, the character itself doesn't change.

$ABCDE \rightarrow CADBE$

$\therefore$ The characters are moved to different positions. This spreads the ~~mis~~information of the plaintext throughout the ciphertext. Hence transposition mainly provides diffusion.

4. Transposition Encryption:-

Encrypt the message "the arrow is pointing north direction". using,

Simple

With key = 4

$\Rightarrow$ Writing row by row into 4 columns, then read column by column

T H E A
 R R O W
 I S P O
 I N T I
 N G N O
 R T H D
 I R E C
 T I O N

CipherText: TRIIMRIT HRSNGTRI IOPTNHEO'
 AWOIODCN

Keyword

with keyword = ATLAS

Order them ' A T L A S
 from left to 1 4 5 3 2 4
 right

T H E A R
 R O W I S
 P O I N T
 I N G N O
 R T H D I
 R E C T I
 O N X X X

CipherText: TRPIRROAINNDTXEWIGHCX RSTOTIXHOONTEN

double

Key 1: B R E A K

Key 2: L O G

Key 1 = BREAK

B R E A K

2 5 3 1 4

T H E A R

R O W I S

P O I N T

I N G N O

R T H D I

R E C T I

O N X X X

Cipher Text: AINNDTXTRPIRROEWIGHCXRSTOIXHOONTEN

Key 2 = L O G

2 3 1

A I N

N D T

A T R P

I R R

O E W

I G H

C R S

T O I

I H O

O N T

E N X

Cipher Text: NTPRWHSTOTXANTIOICTIOEIDRREGROHNN

Decryption:

⇒ Simple Transposition with Key = 10.

Given,

Cipher text:

TIDIE MRNME REOGO SANOW RSLTP FEEDO SSSNU
AHTUD BIENP GODAE PEIXD ELNSX

Remove Spaces:

TIDIEMRNMEREOGOSANOWRSLTPEEEDOSSSNUAHTUDBIENPGODAEPE
IXDELNSX

Total number of characters = 60

Key = 10

No. of. columns = 10

No. of. rows = $60/10 = 6$.

Divide ciphertext into 10 groups, each containing 6 letters:

T I D I E M

R N M E R E

O G O S A N

O W R S L T

P E E E D O

S S S N U A

H T U D B I

E N P G O D

A E P E I X

D E L N S X

Write these as columns (or) read column-wise and you get:

Plain Text: TROOPS HEADING WEST NEED MORE
SUPPLIES SEND GENERAL DUBOIS MEN TO AID.

→ Keyword → PLAIN | CIPHER:
VEY EAX ARA ATX HGD

— PLAIN

5 3 1 2 4 → Total - 15

→ length = $15/5 = 3$

Col 1 = V E Y

Col 2 = E A X

Col 3 = A R A

Col 4 = A T X

Col 5 = H G D

P L A I N

5 3 1 2 4

H A V E A

G R E A T

D A Y X X

Plain Text: HAVE A GREAT DAY

5. Permutation

Find the inverse permutation for —

(1) 4. 3 1 5 2

$P = (4, 3, 1, 5, 2)$

1 is in position 3

similarly, 2 → 5

3 → 2

4 → 1

5 → 4

Inverse Permutation = (3, 5, 2, 1, 4)

(2) 7 4 1 3 5 2 6

$P = (7, 4, 1, 3, 5, 2, 6)$

1 → 3

5 → 5

2 → 4

6 → 7

3 → 1

7 → 1

4 → 2

$P^{-1} = (3, 4, 1, 2, 5, 7, 1)$

(3) 4 1 3 2

$$P = (4, 1, 3, 2)$$

$$1 \rightarrow 2$$

$$2 \rightarrow 4$$

$$3 \rightarrow 3$$

$$4 \rightarrow 1$$

$$P^{-1} = (2, 4, 3, 1)$$

(4) 3 5 1 4 2

$$P = (3, 5, 1, 4, 2)$$

$$1 \rightarrow 3$$

$$2 \rightarrow 5$$

$$3 \rightarrow 1$$

$$4 \rightarrow 4$$

$$5 \rightarrow 2$$

$$P^{-1} = (3, 5, 1, 4, 2)$$

→ Keyword → SECRET

~~SECRET~~

Total → 36

Length → 6

Col 1 = I L E D T N
Col 2 = H P T R S O
Col 3 = I I T M A X
Col 4 = S A X E R X
Col 5 = T A N O N I
Col 6 = S N F O T X

S E C R E T
5 2 1 4 3 6
T H I S I S
A P L A I N
N T E X T F
O R D E M O
N S T R A T
I O N X X X

Read Column by Column;

Plain Text: THIS IS A PLAIN TEXT FOR
DEMONSTRATION.